BITS Pilani

Database Design and Applications

CSIZG518/SEZG518/SSZG518

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innovate achieve lead

BITS Pilani

<CSIZG518/SEZG518/SSZG518, DBDA>
Lecture No:04

Degree of a relationship

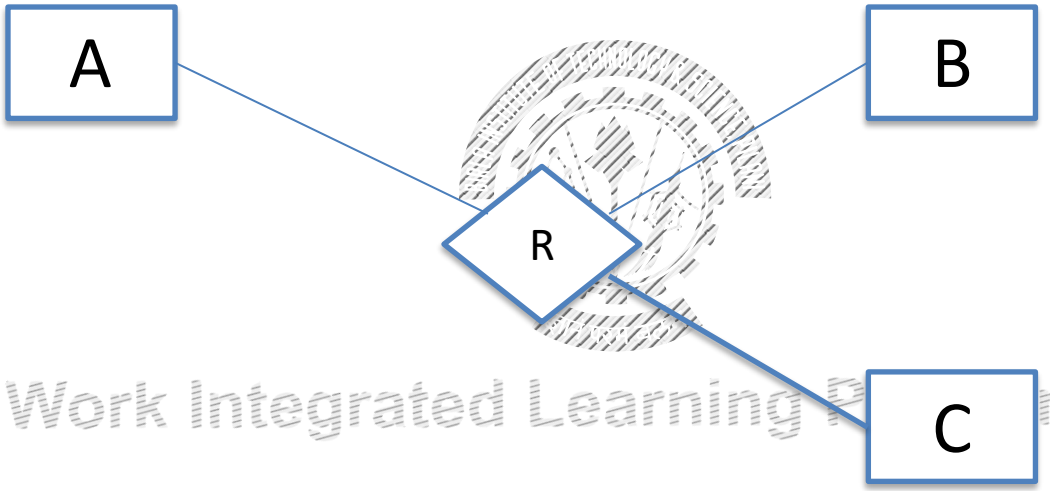
Degree the number of participating entities

- Degree 2: Binary
- Degree 3: Ternary
- Degree 4: n-ary

Binary relationships are very common and widely used

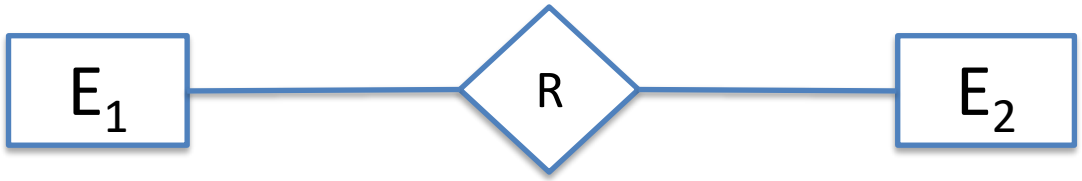
Diagrammatic Notation of Relationships

Relationship- diamond shaped box



Rectangle of each participating entity is connected by a line to this diamond.
Name of the relationship is written in the box

Binary Relationships & Cardinality Ratio



The Maximum number of entities from E_2 that an entity from E_1 can possibly be associated thru R (and vice-versa).Determines the cardinality ratio of R.

Four Possibilities are usually specified.

1. One-to-One (1:1)
2. One-to-Many (1:N)
3. Many-to-Many (M:N)
4. Many-to-One (M:1)

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Cardinality Ratio

One-to-One

An E_1 entity may be associated with at most one E_2 entity and similarly an E_2 entity may be associated with at most one E_1 entity.

One-to-many

An E_1 entity may be associated with many E_2 entities whereas an E_2 entity may be associated with at most one E_1 entity.

Many-to-One

An E_2 entity may be associated with many E_1 entities where as an E_1 entity may be associated with at most one E_2 entity

Many-to-Many

Many E_1 entities may be associated with a single E_2. Entity E_2 entity and a single E_1 entity may be associated with many E_2 entities.

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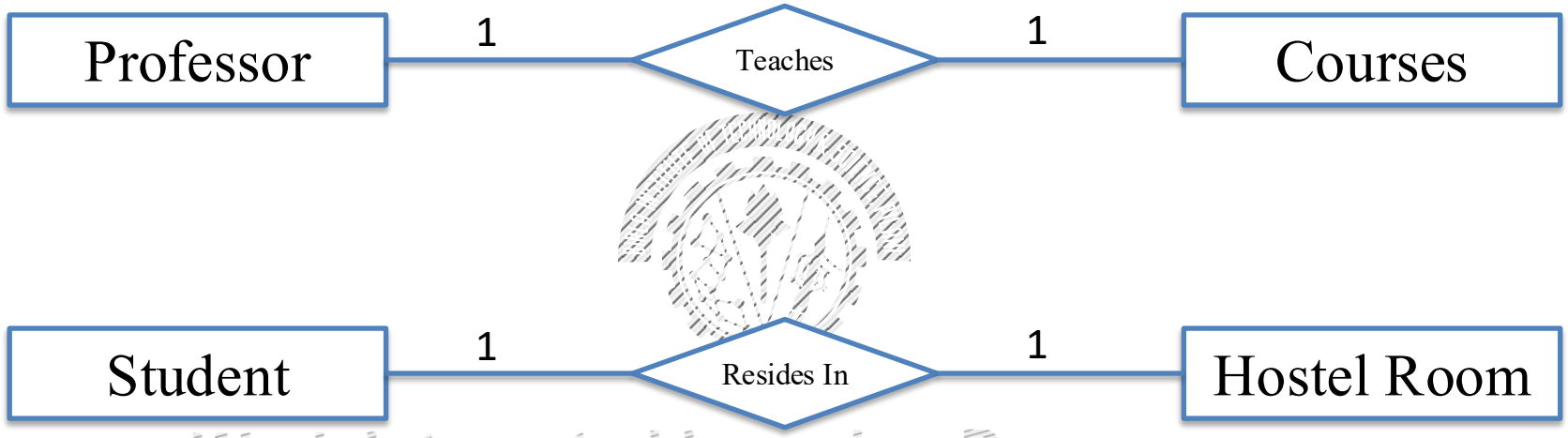
Cardinality Ratio (Example)

One-to-one



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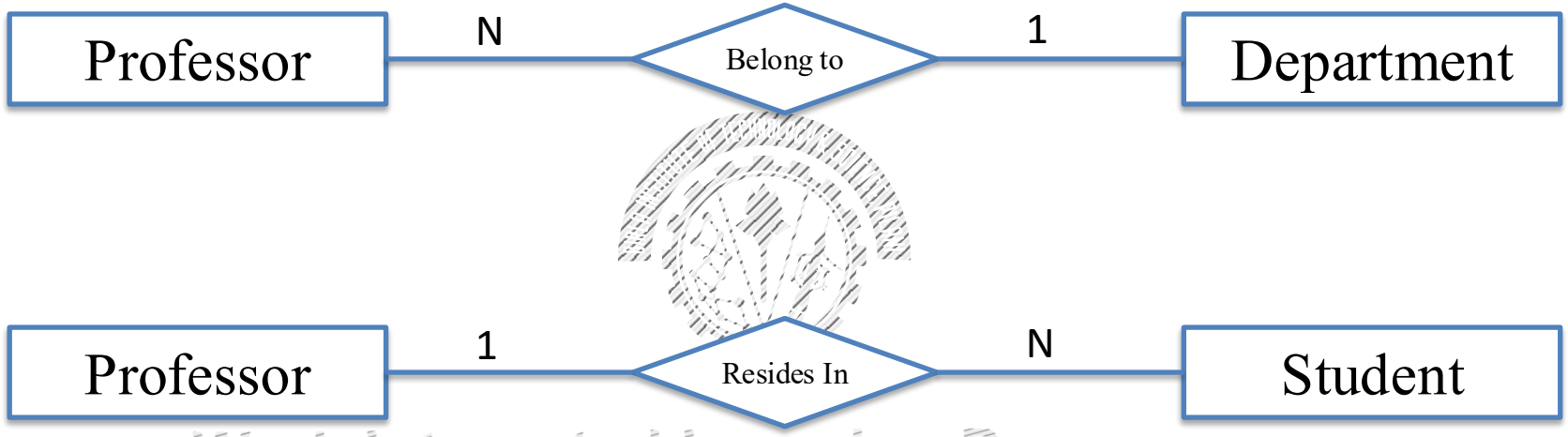
Cardinality Ratio (Example)

Many-to-One /One-to-Many



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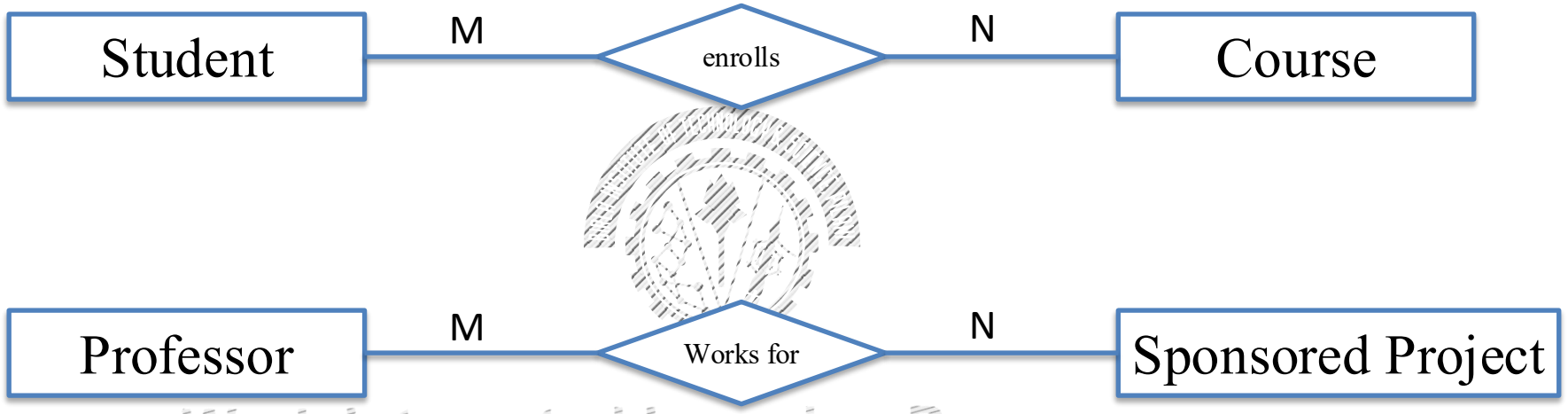
Cardinality Ratio (Example)
Many-to-Many



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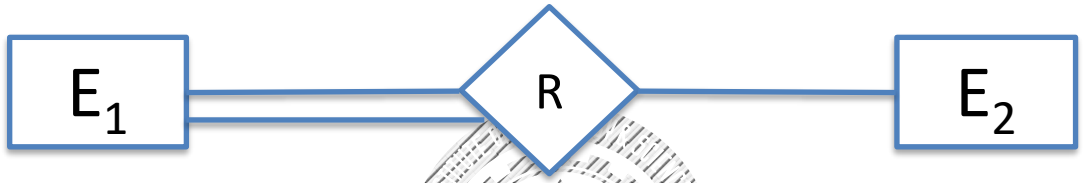
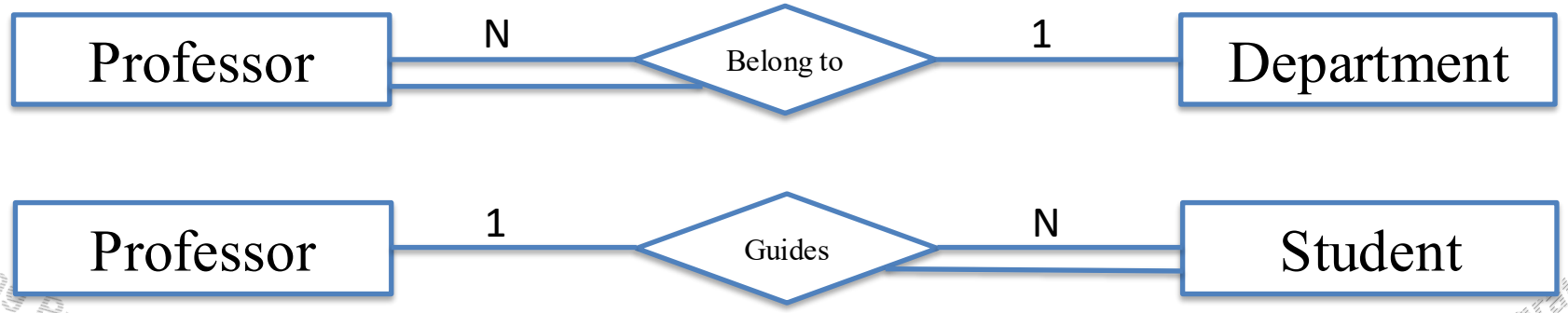
Participation Constraints

An Entity set may participate in a relation either totally or partially



Total participation: every entity in the set is involved in some association of the relationship.

Partial Participation: Not all entities in the set are involved in association of the relationship.



Structural Constraints

- Cardinality Ratio & Participation constraints are together called **structural constraints**
- They are called constraints as the data must satisfy them to be consistent with the relationship.
- They are called constraints as the data must satisfy them to be consistent with the requirements.
- min-max notation: pair of numbers (m,n) placed on this line connecting an entity to the relationship.

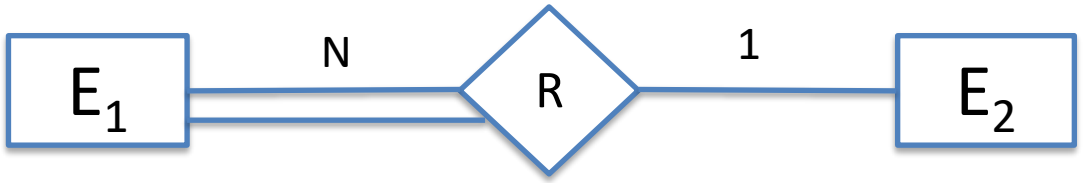
  m: the minimum number of times a particular entity must appear in the relationship tuples at any point of time.

  0 → Partial Participation

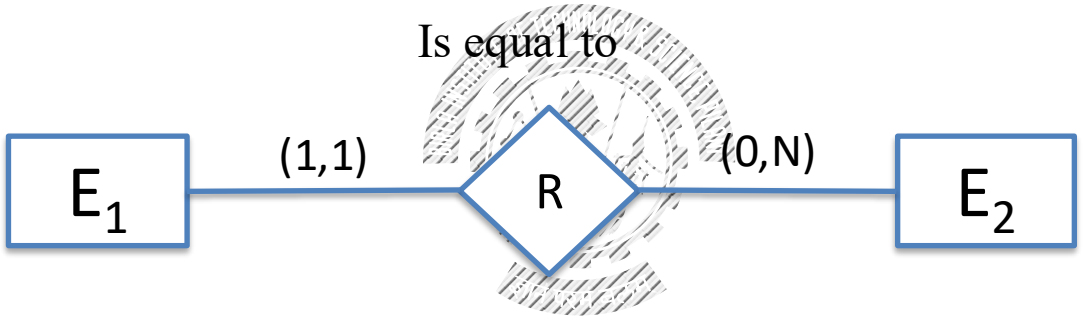
  ≥ → Total Participation

  n: Similarly the maximum numbers of times a particular entity can appear int eh relationship tuples at any point of time.

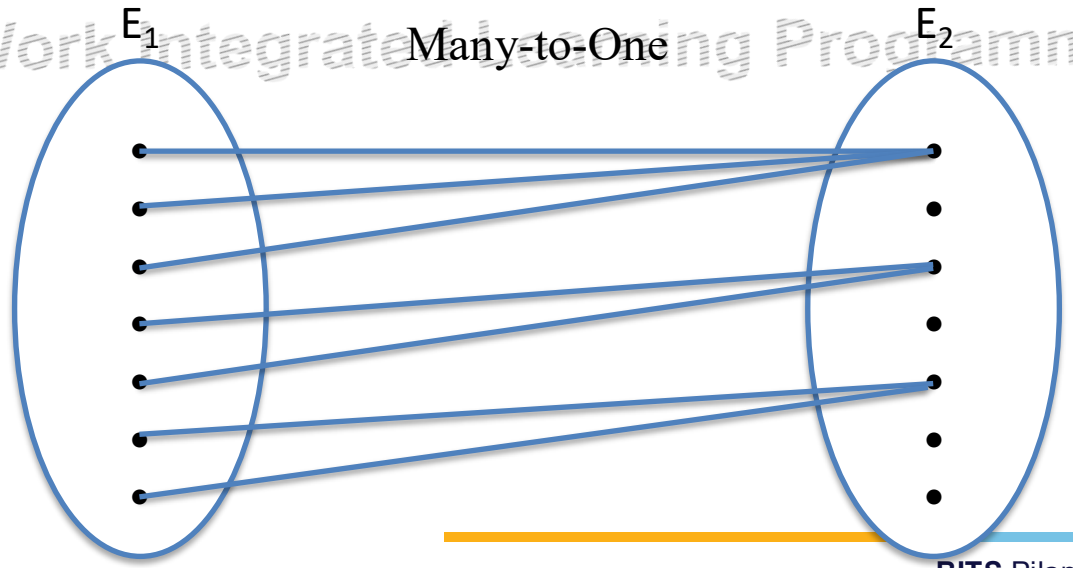
Comparing the Notation



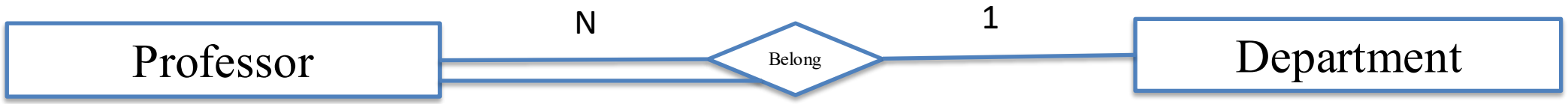
Is equal to



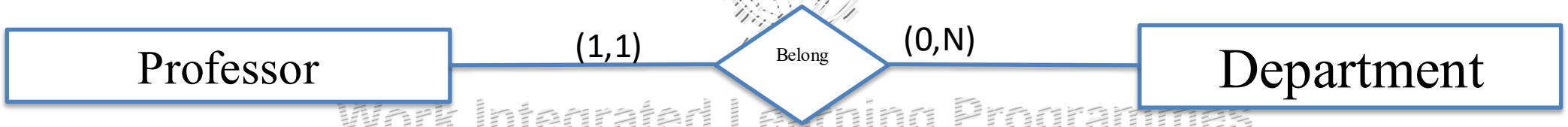
Many-to-One



Example



Is equal to



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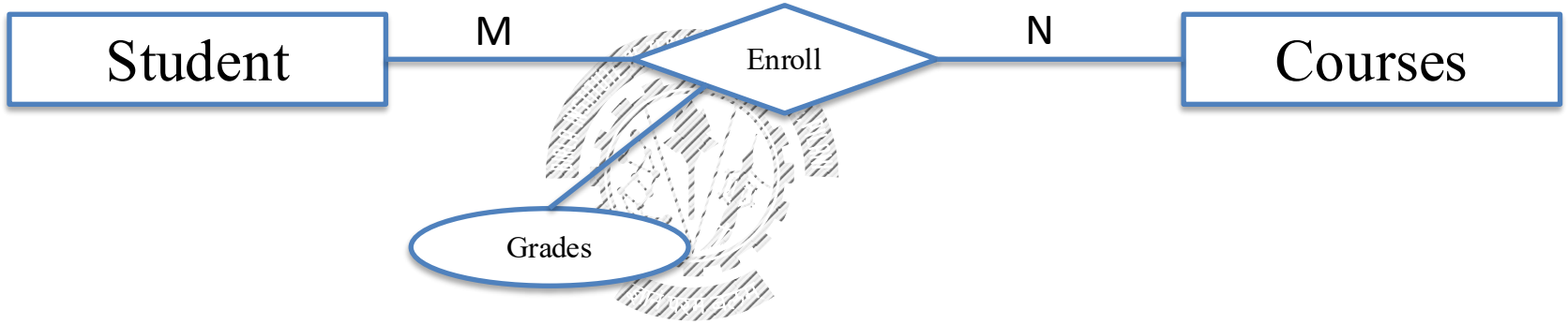
Attributes for Relationship Types

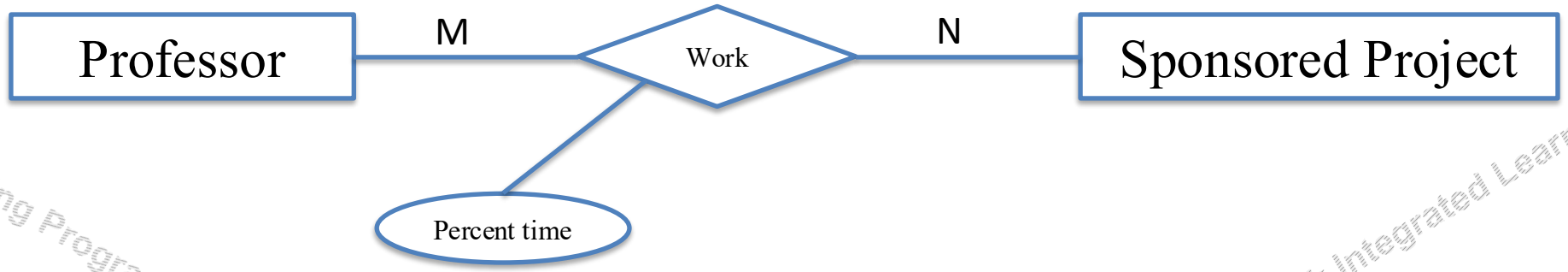
Relationship types can also have attributes

Properties of the association of entities



Grade gives the letter grades (A,A$^-$,B,B$^-$,….) earned by the student for a course

**neither an attributes of student nor that of course



Recursive Relationships & Role Name

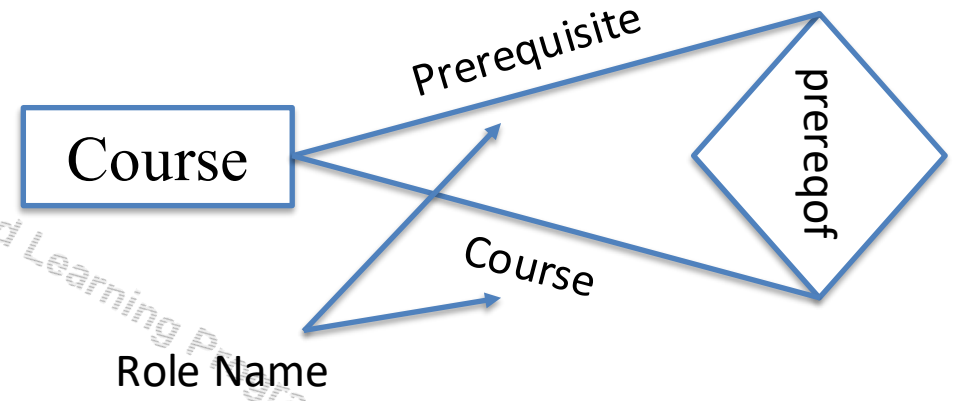
Recursive relationship: An entity set relating to itself gives rise to a recursive relationship .

Eg: The relationship prereqof is an example of a recursive relationship on the entity course.

Role name: Used to specify the exact role in which the entity participates in the relationships

Essential in case of recursive relationship

Can be optionally specified in non-recursive cases



Complete Example for E/R Schema: Specifications

In an educational institute, there are several departments and each student belongs to one of them.
Each department has a unique department number, a name, a location, phone number and is headed by a professor.
Professors have a unique employee Id, name and a phone number. A professor works for exactly one department.

We like to keep track of the following details regarding students: name, unique roll number, sex, phone number, date of birth, age and one or more email addresses. Students have a local address consisting of the hostel name and the room number. They also have home address consisting of house number, street, city and PIN.
It is assumed that all students reside in the hostels.

A course taught in a semester of the year is called a section. There can be several sections of the same course in a semester; these are identified by the section number. Each section is taught by a professor and has its own timings and a room to meet. students enroll for several sections in a semester.

Each course has a name, number of credits and the department that offers it. A course may have other courses as pre-requisites i.e., courses to be completed before it can be enrolled in.

Professors also undertake research projects. These are sponsored by funding agencies and have a specific start date, end date and amount of money given. More than one professor can be involved in a project. Also, a professor may be simultaneously working on several projects. A project has a unique projectID.

Solution

Refer to the live lecture

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Thank You

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